RAY SUMMIT 2024

Connecting idle GPUs worldwide to fulfill part of the infinite demand for AI Computation

Aline Almeida, IO.net

Intro

IO.NET

Aline Almeida

Germany

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linkedin.com/in/alinealmeida

IO.NET

IOG | FOUNDATION

Partners SOLANA APTOS Render Network Filecoin

Agenda

- **About IO.net**
  - The world's largest decentralized GPU cloud
  - "Airbnb" of GPUs
- **Characteristics**
  - Decentralized, Heterogeneous, Geographically distributed nodes
- **Measuring Node Reliability**
- **Ray.io to Enable GPU Clusters for AI Scaling**
  - Customization: Dashboard and Authentication
  - Demo - Distributed Training with 56 GPUs
- **Ongoing Optimization Efforts**
- **Research Partnership**

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Underutilized Supply meets AI Demand

IO.NET

Independent Data Centers

Crypto Mining Farms

Consumer GPUs

IO.NET | CLOUD

Ray

Deploy a Decentralized Ray Cluster

Mega-Ray

Deploy a Decentralized Mega-Ray Cluster

...others

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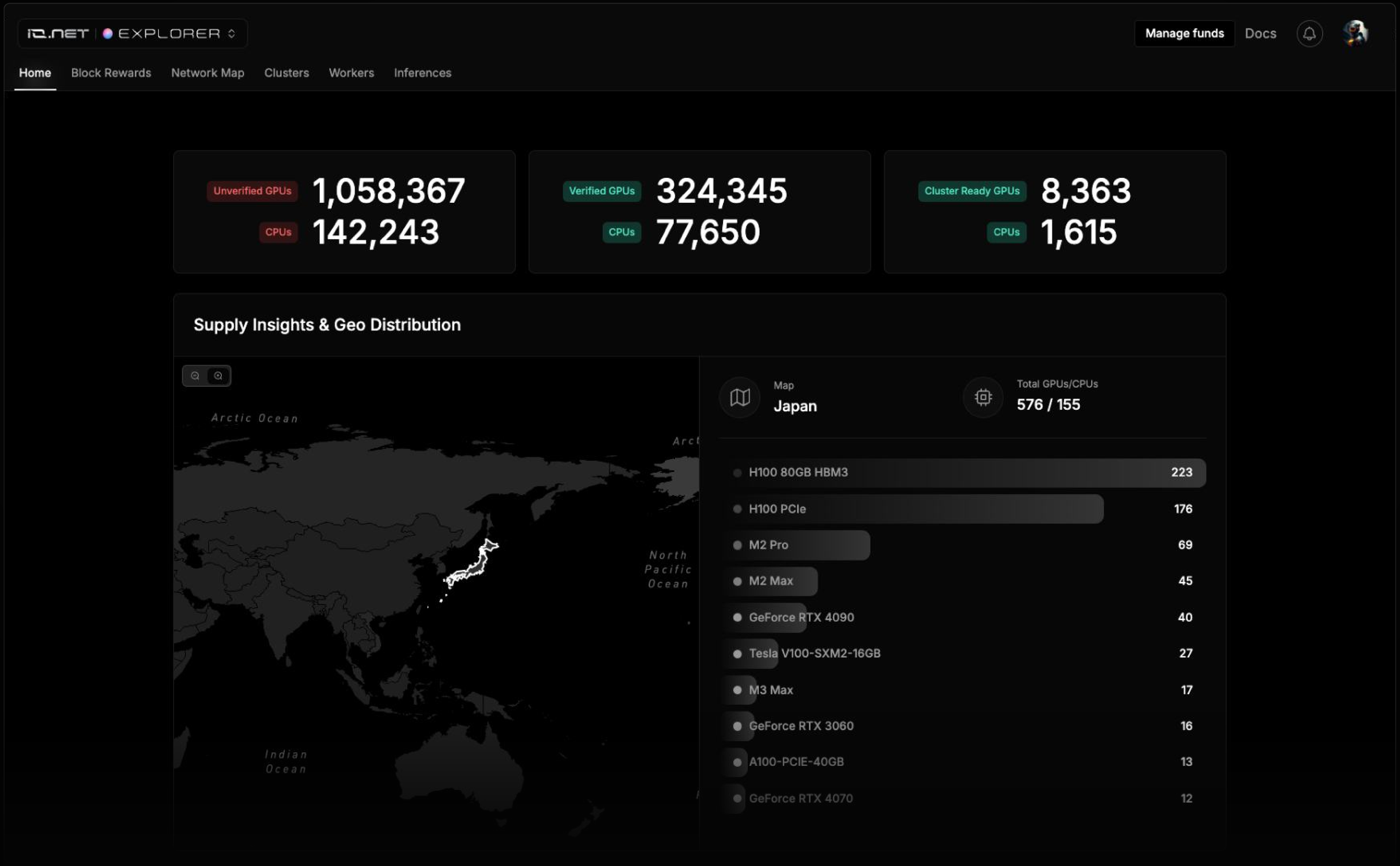

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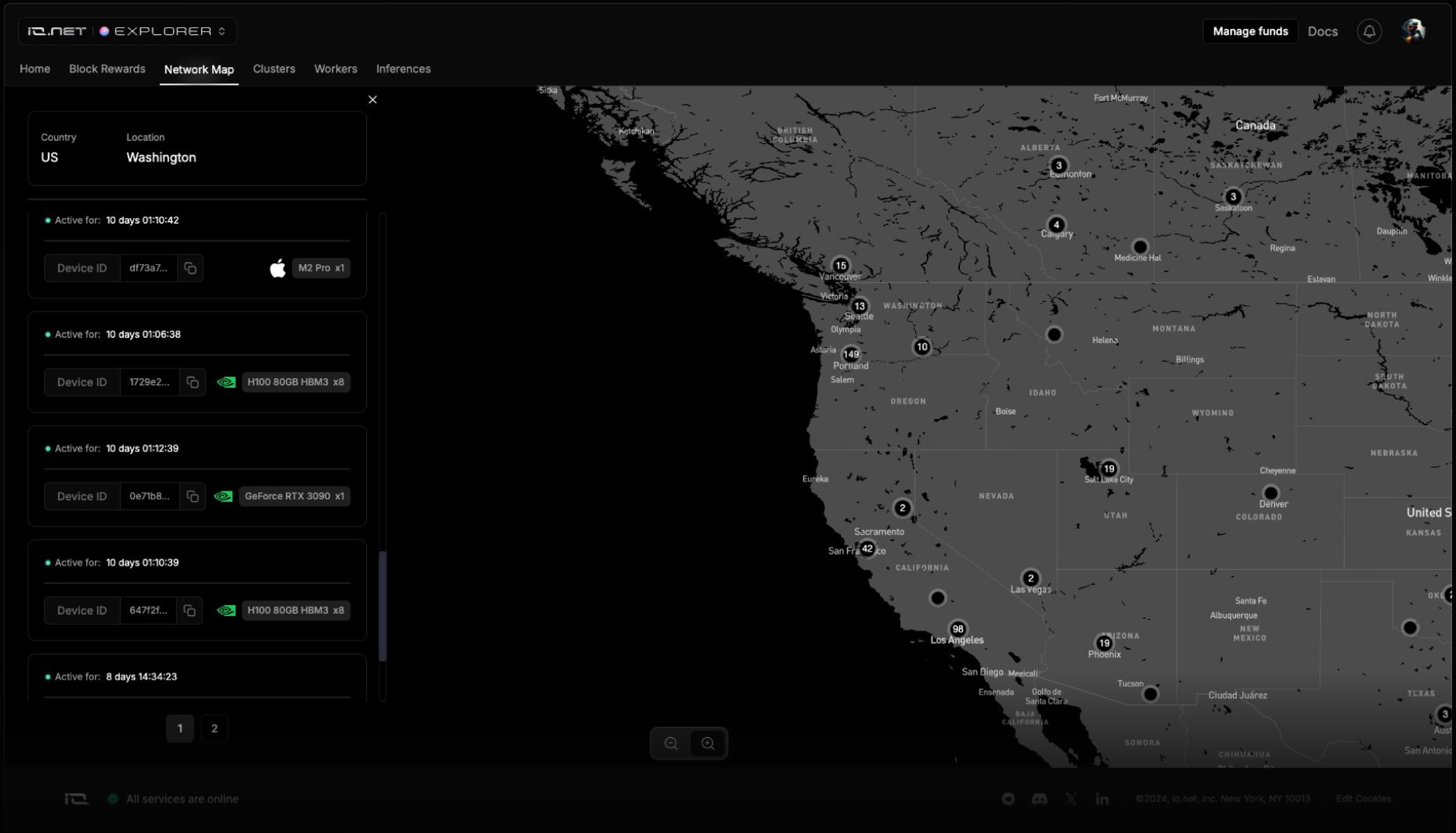

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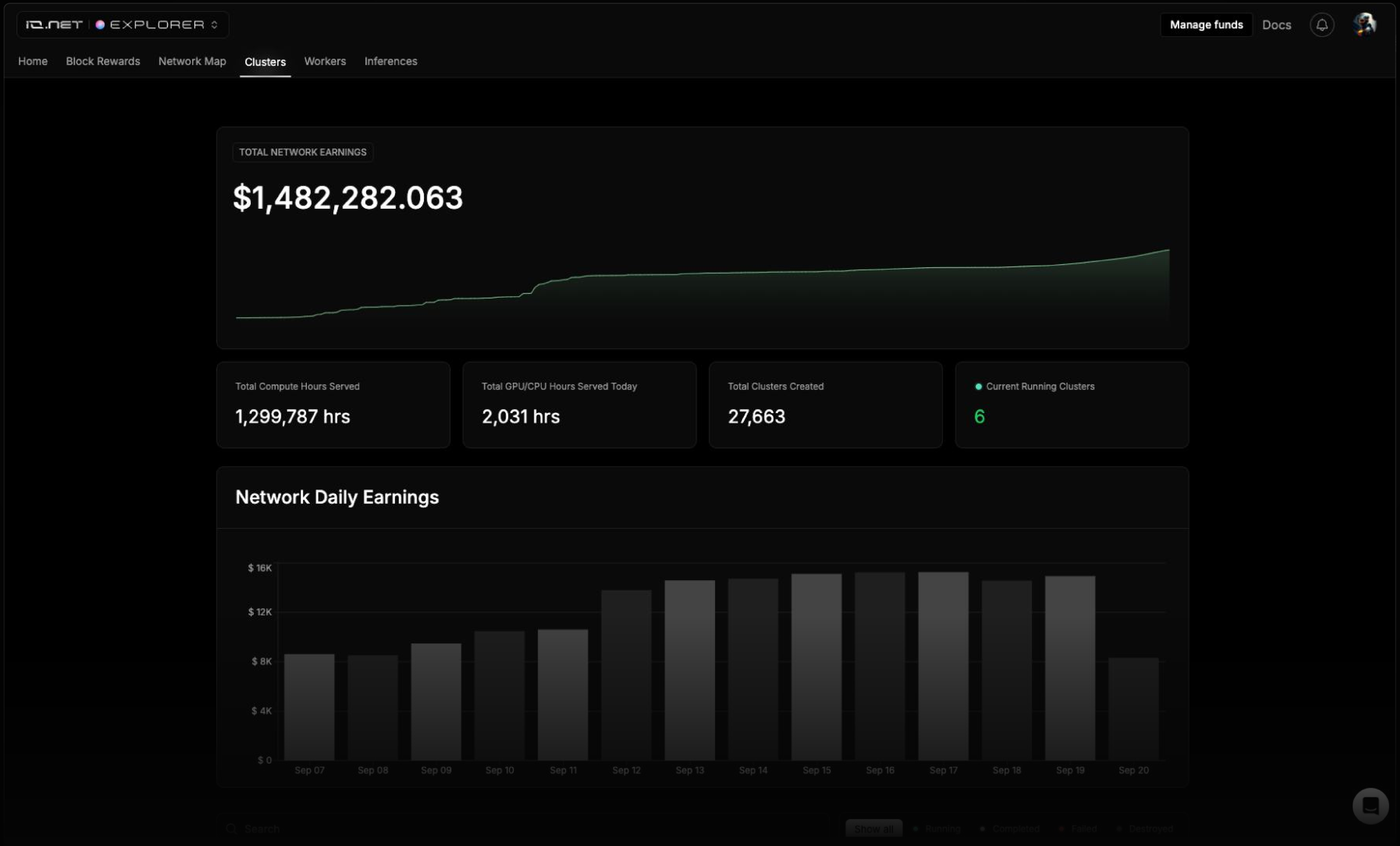

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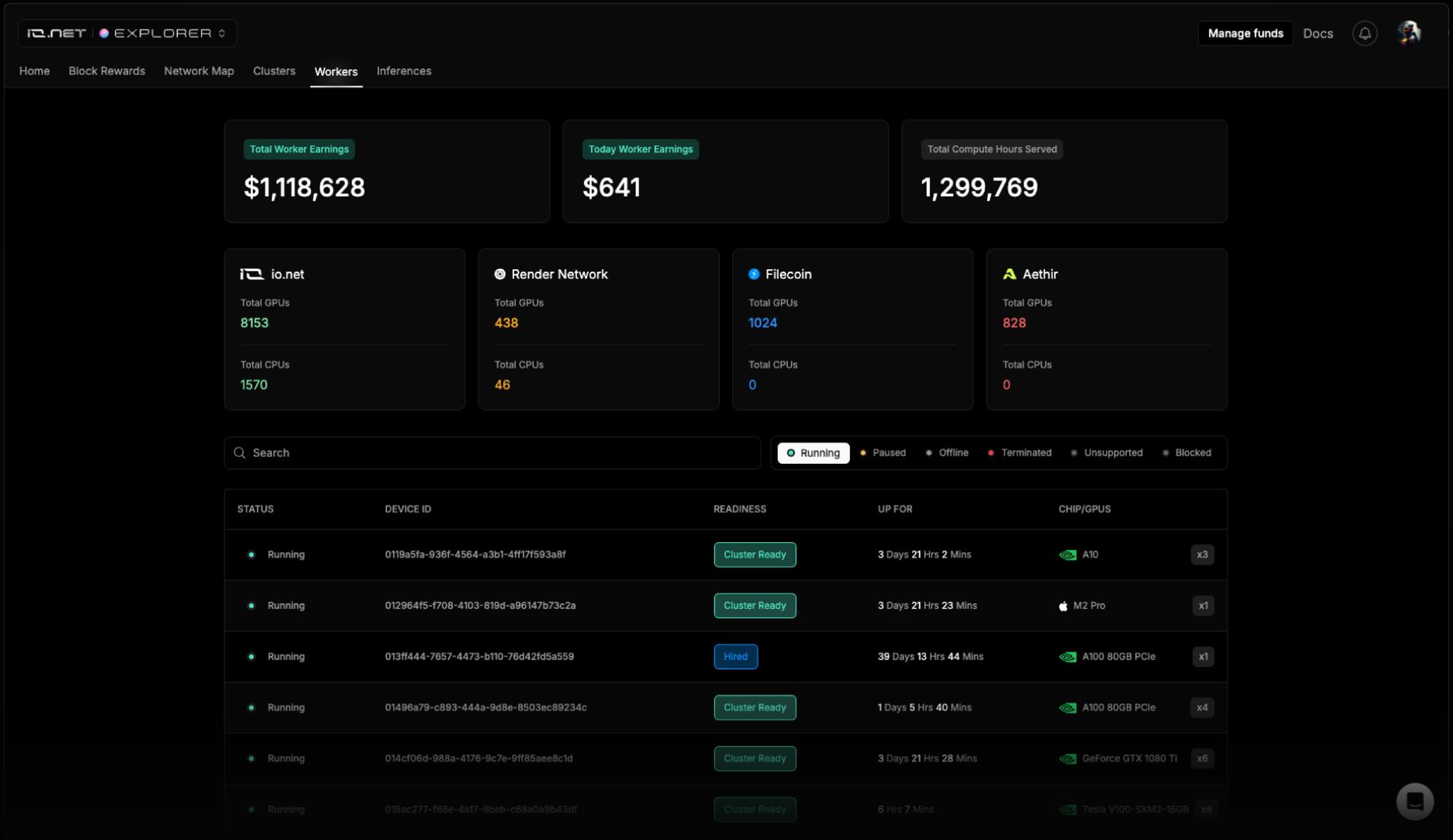

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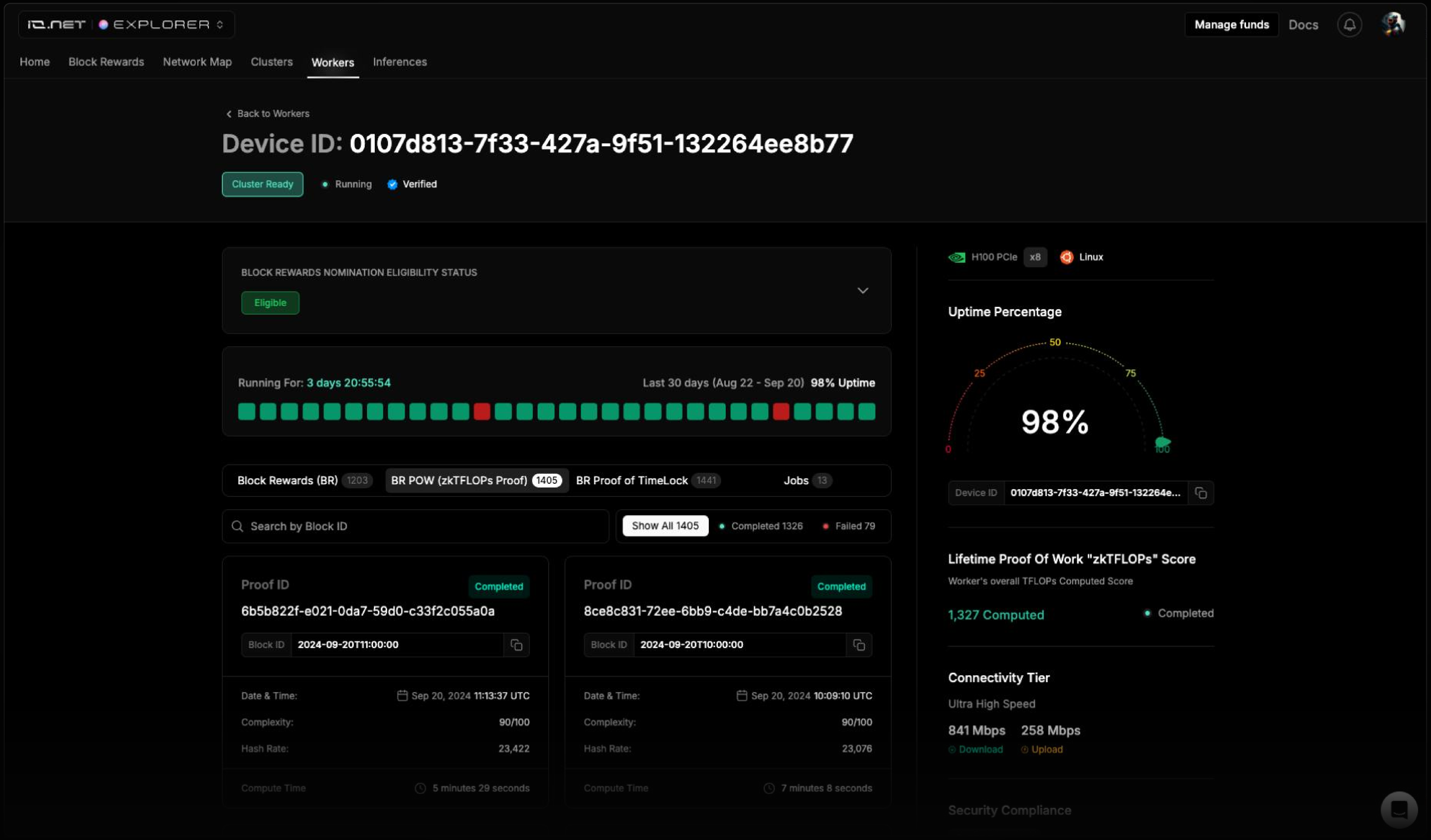

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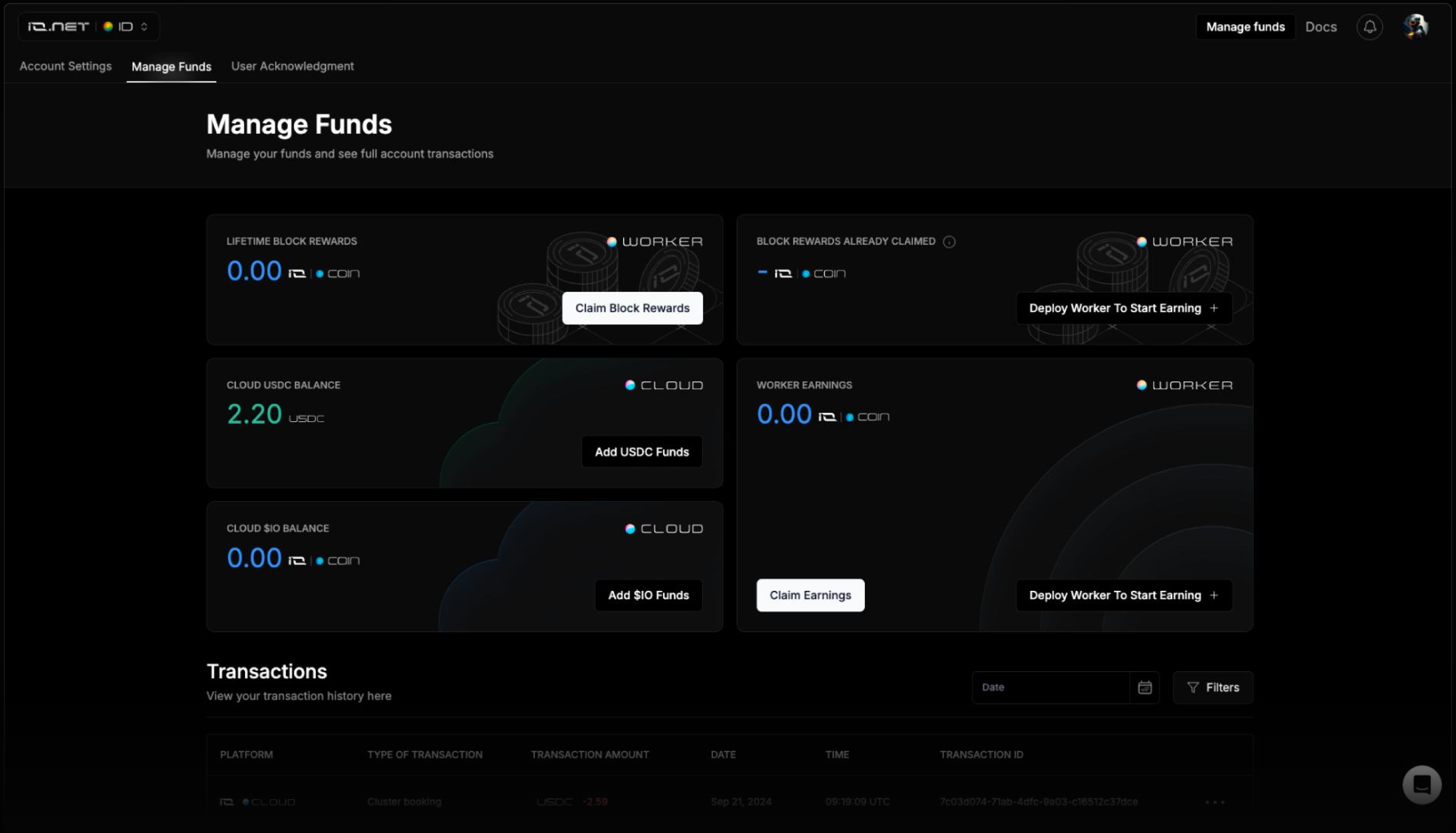

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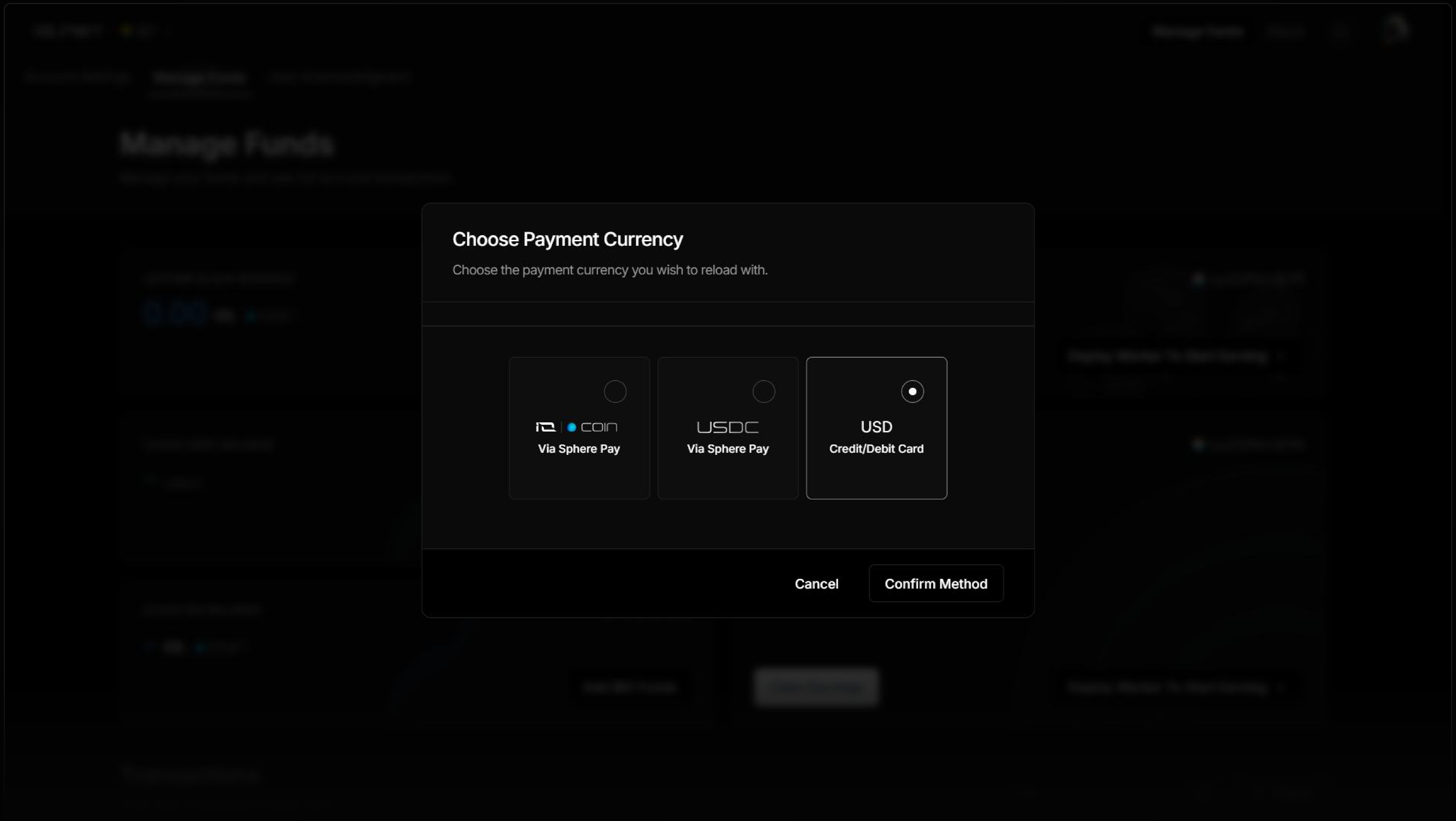

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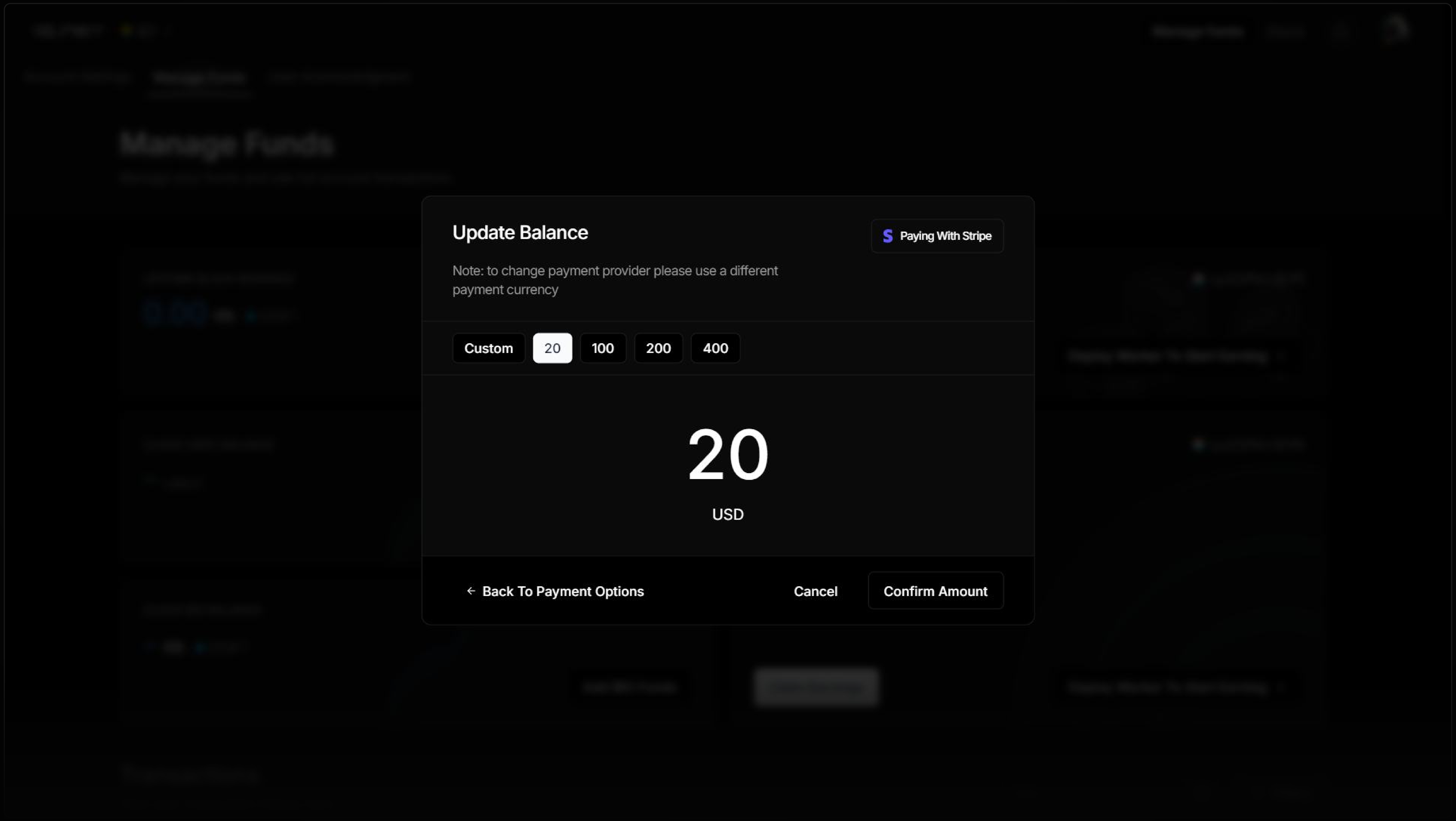

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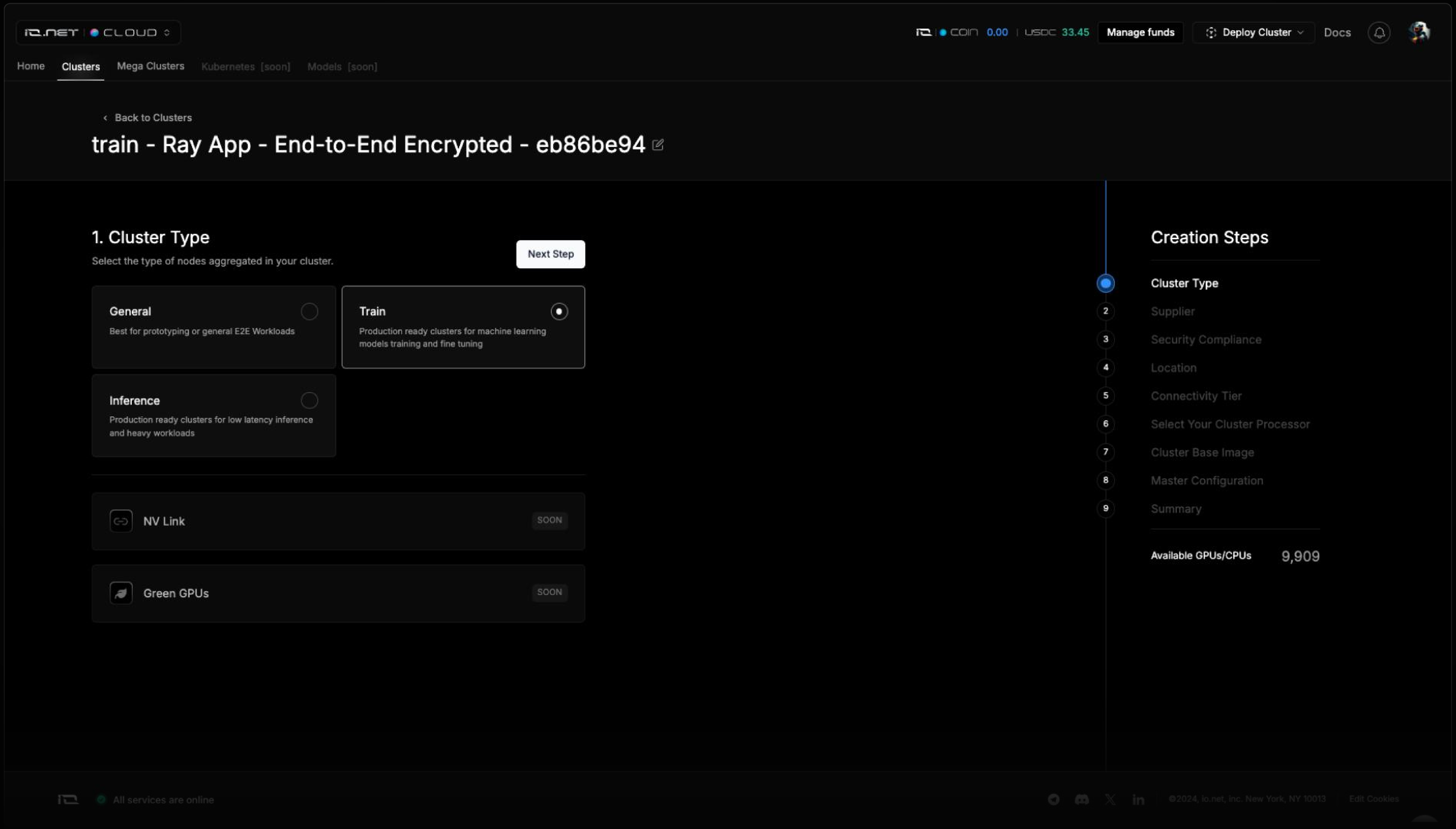

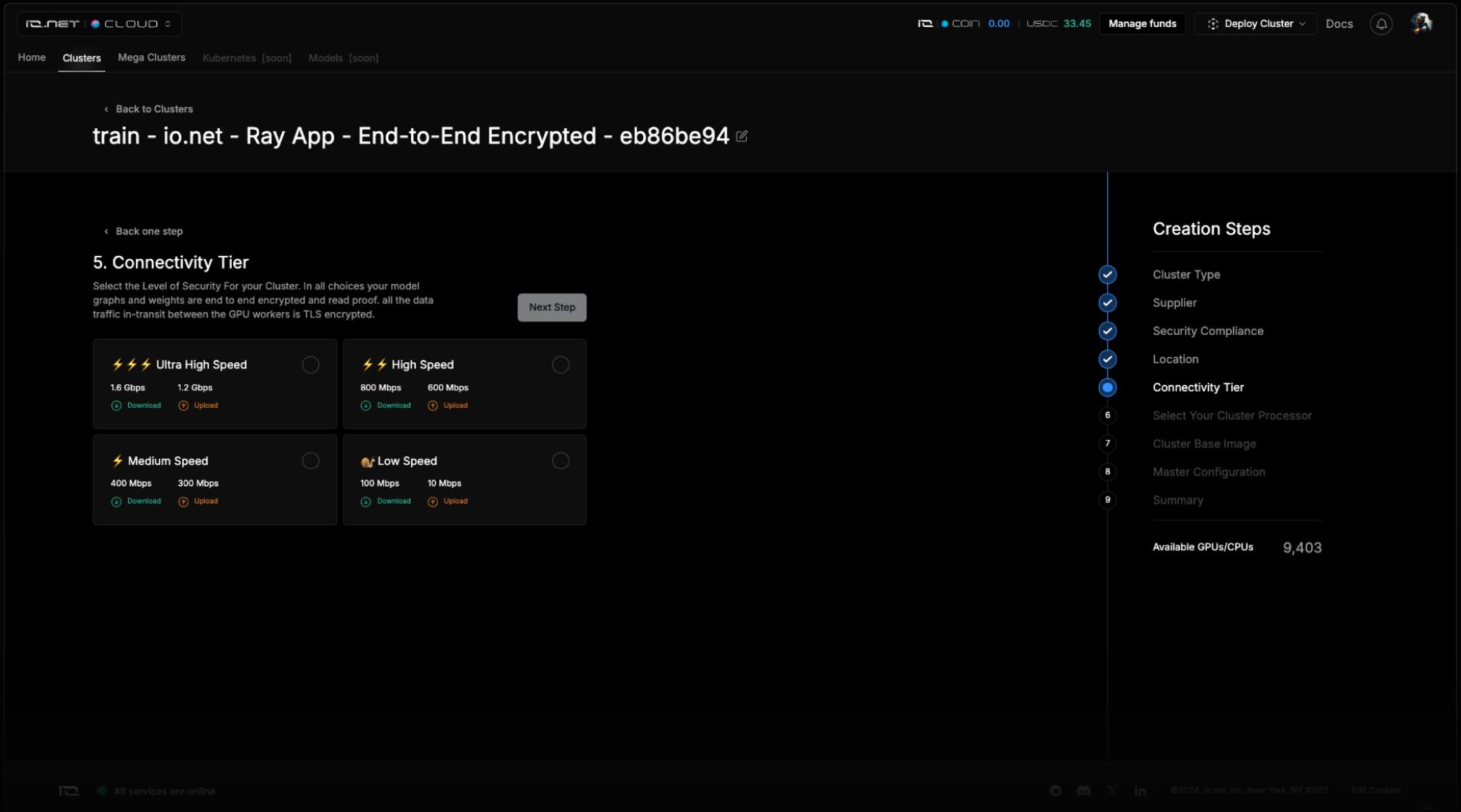
CLOUD
COIN 0.00
USDC 33.45
Manage funds
Deploy Cluster
Docs
Home
Clusters
Mega Clusters
Kubernetes [soon]
Models [soon]
Back to Clusters
train - io.net - Ray App - End-to-End Encrypted - eb86be94
Back one step
5. Connectivity Tier
Select the Level of Security For your Cluster. In all choices your model graphs and weights are end to end encrypted and read proof. all the data traffic in-transit between the GPU workers is TLS encrypted.
Next Step
Ultra High Speed
1.6 Gbps
1.2 Gbps
Download
Upload
High Speed
800 Mbps
600 Mbps
Download
Upload
Medium Speed
400 Mbps
300 Mbps
Download
Upload
Low Speed
100 Mbps
10 Mbps
Download
Upload
Creation Steps
Cluster Type
Supplier
Security Compliance
Location
Connectivity Tier
6 Select Your Cluster Processor
7 Cluster Base Image
8 Master Configuration
9 Summary
Available GPUs/CPUs 9,403

RAY
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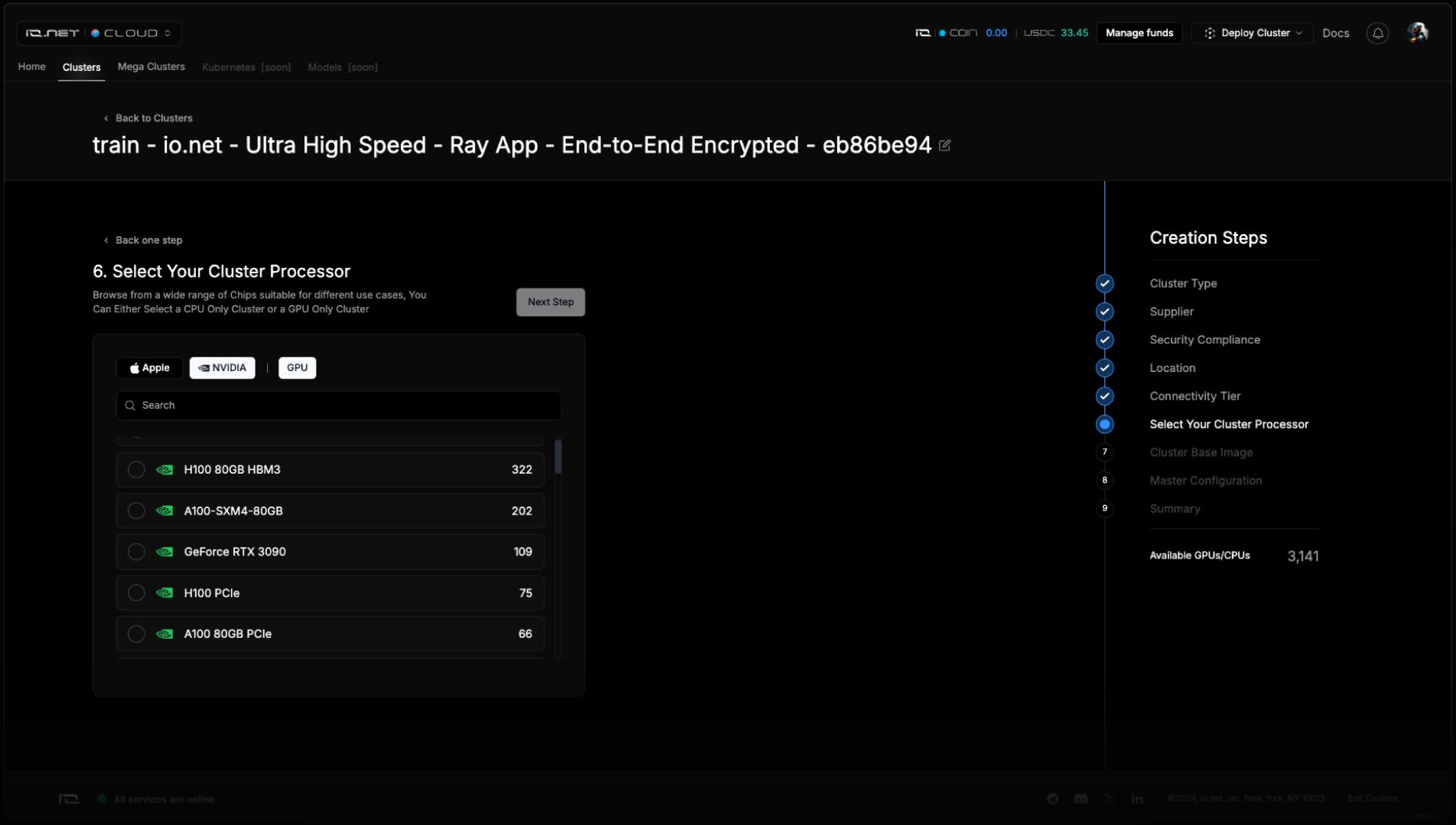
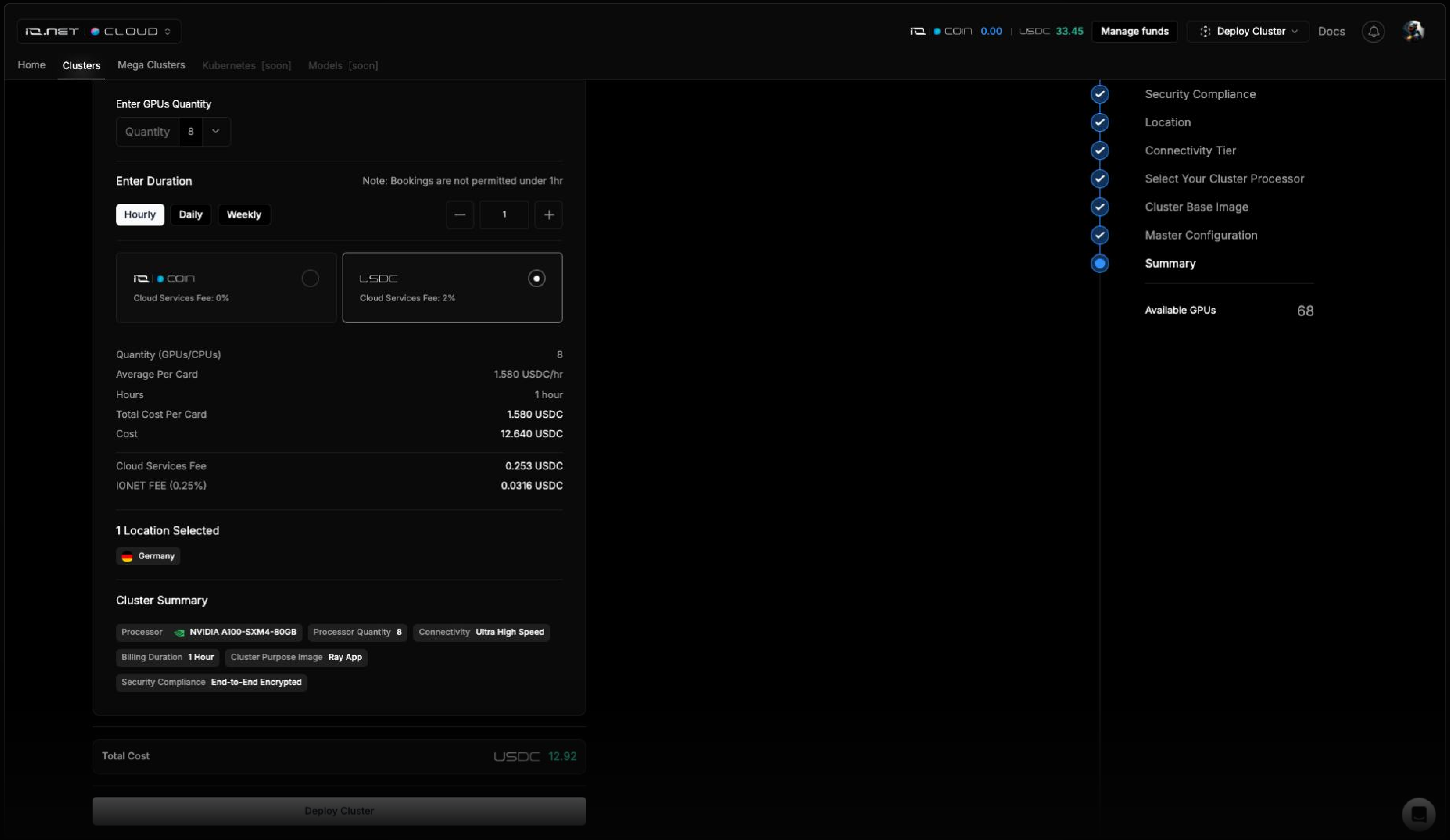
2024




CLOUD
COIN 0.00
USDC 33.45
Manage funds
Deploy Cluster
Docs
Home
Clusters
Mega Clusters
Kubernetes [soon]
Models [soon]
Enter GPUs Quantity
Quantity 8
Enter Duration
Note: Bookings are not permitted under 1hr
Hourly
Daily
Weekly
1
COIN
Cloud Services Fee: 0%
USDC
Cloud Services Fee: 2%
Quantity (GPUs/CPUs) 8
Average Per Card 1.580 USDC/hr
Hours 1 hour
Total Cost Per Card 1.580 USDC
Cost 12.640 USDC
Cloud Services Fee 0.253 USDC
IONET FEE (0.25%) 0.0316 USDC
1 Location Selected
Germany
Cluster Summary
Processor NVIDIA A100-SXM4-80GB
Processor Quantity 8
Connectivity Ultra High Speed
Billing Duration 1 Hour
Cluster Purpose Image Ray App
Security Compliance End-to-End Encrypted
Total Cost
USDC 12.92
Deploy Cluster
Security Compliance
Location
Connectivity Tier
Select Your Cluster Processor
Cluster Base Image
Master Configuration
Summary
Available GPUs 68

RAY
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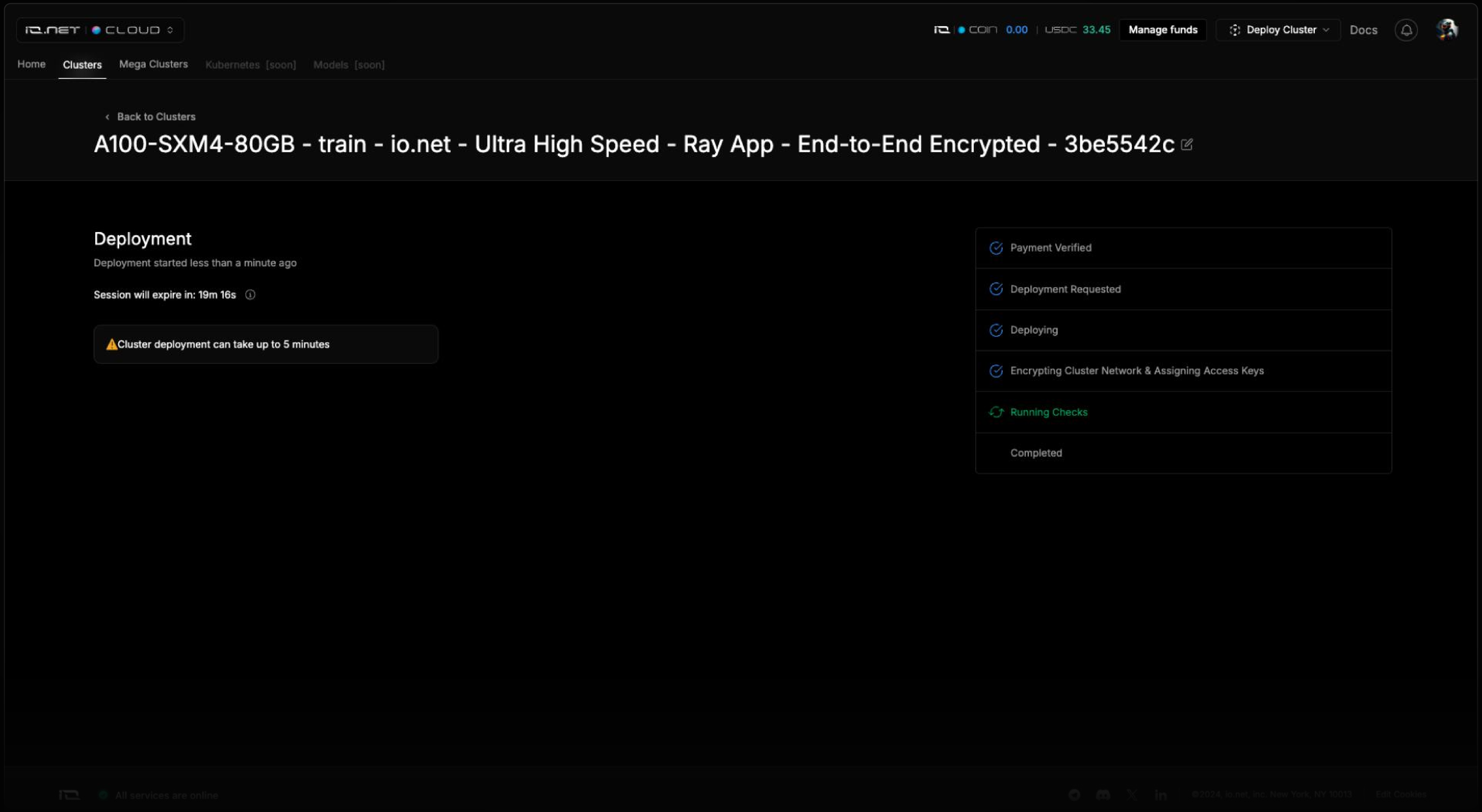
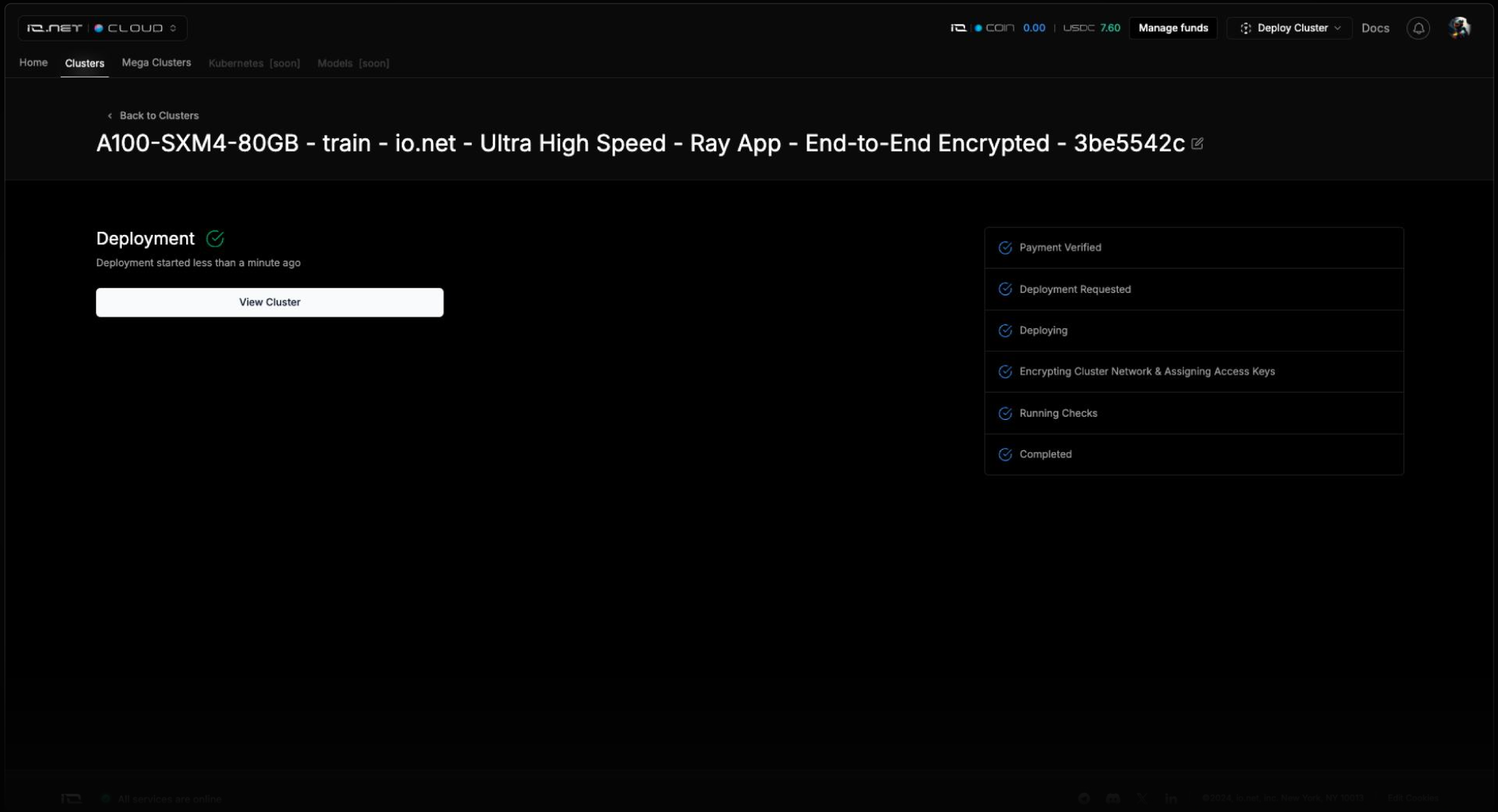
2024




CLOUD
COIN 0.00
USDC 7.60
Manage funds
Deploy Cluster
Docs
Home
Clusters
Mega Clusters
Kubernetes [soon]
Models [soon]
Back to Clusters
A100-SXM4-80GB - train - io.net - Ultra High Speed - Ray App - End-to-End Encrypted - 3be5542c
Deployment
Deployment started less than a minute ago
View Cluster
Payment Verified
Deployment Requested
Deploying
Encrypting Cluster Network & Assigning Access Keys
Running Checks
Completed

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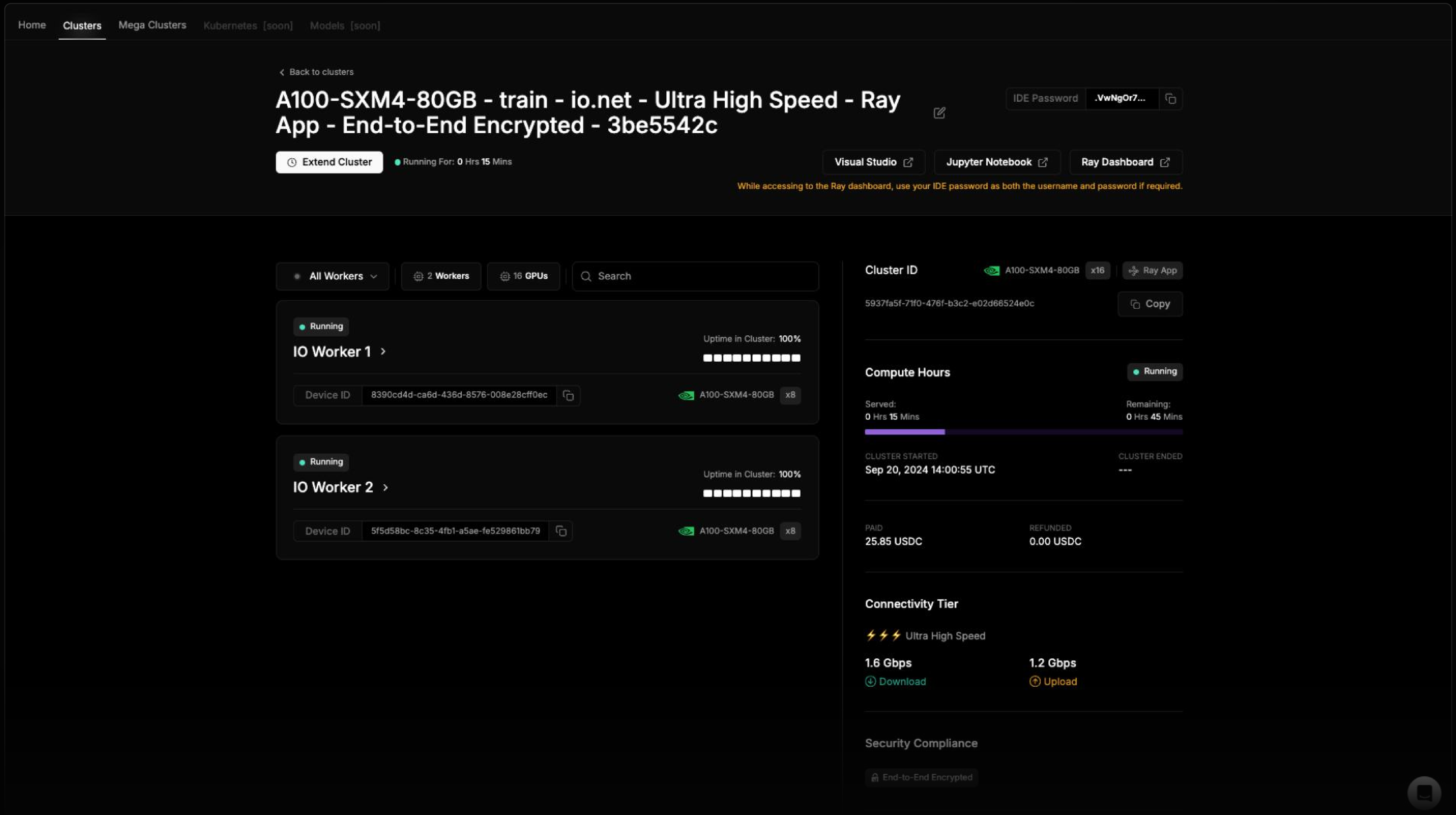

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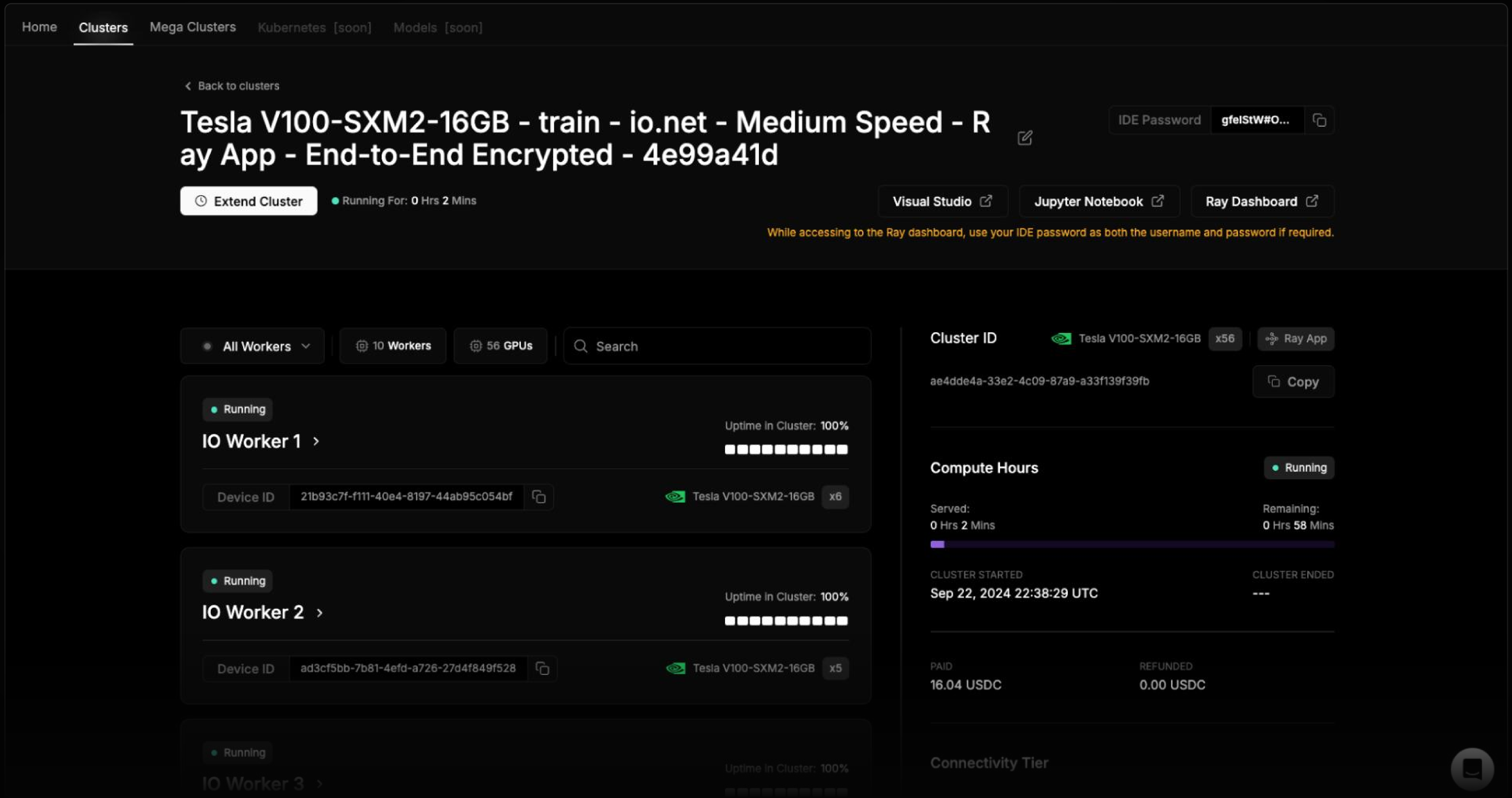
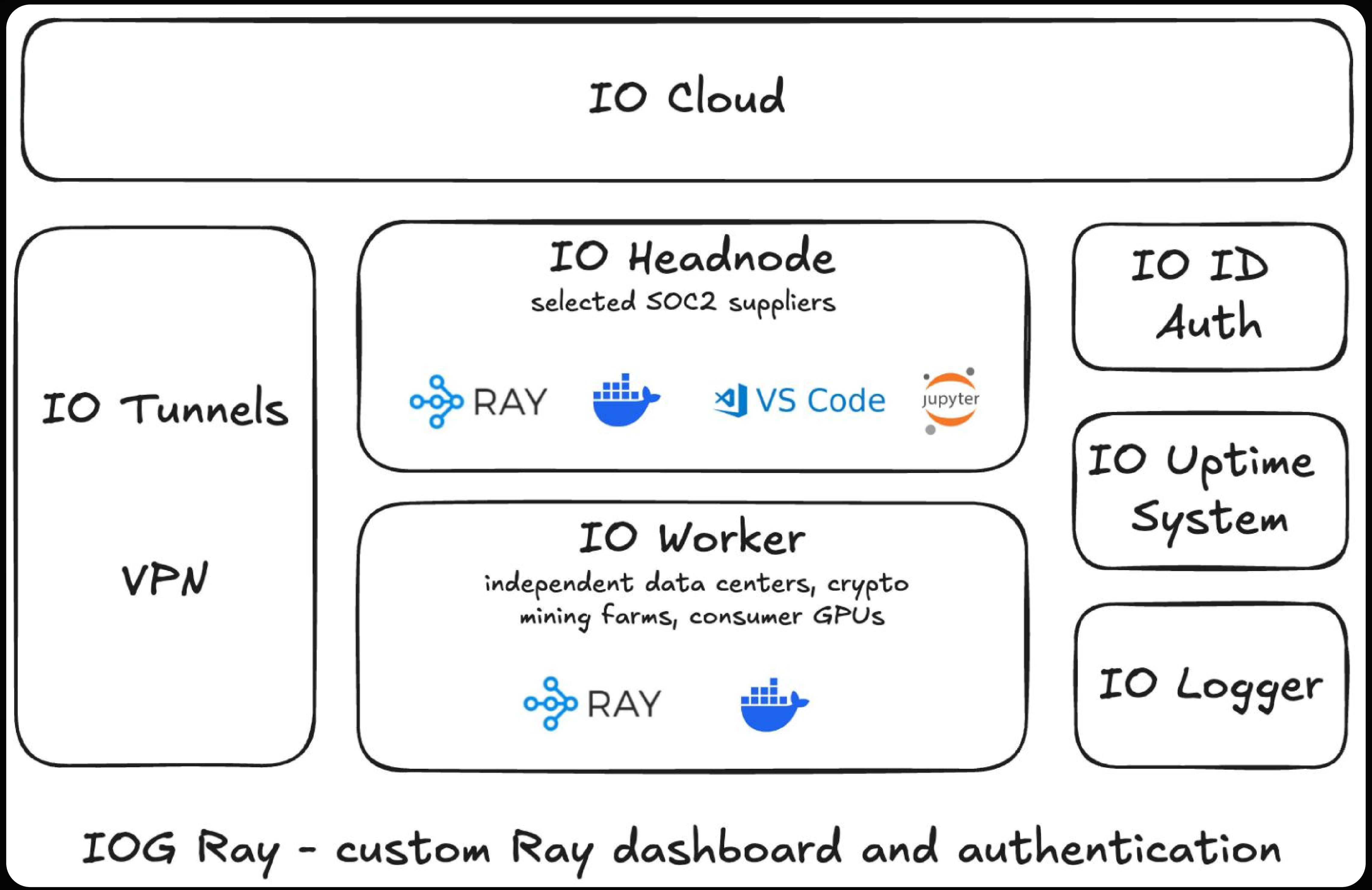


Ray Cluster at IO.net



jupyter ray-summit-demo Last Checkpoint: 57 seconds ago

File Edit View Run Kernel Settings Help   Trusted

Code   Open in...   Python 3 (ipykernel)

[1]:
```python
import ray
```

```
2024-09-22 15:42:25,331 INFO util.py:154 -- Outdated packages:
  ipywidgets==7.7.2 found, needs ipywidgets>=8
Run `pip install -U ipywidgets`, then restart the notebook server for rich notebook output.
```

[2]:
```python
ray.init()
```

```
2024-09-22 15:42:26,968 INFO worker.py:1585 -- Connecting to existing Ray cluster at address: 100.115.112.62:6379...
2024-09-22 15:42:26,980 INFO worker.py:1761 -- Connected to Ray cluster. View the dashboard at 127.0.0.1:8265
```

[2]:

RAY

Python version:	3.9.19
Ray version:	2.30.0
Dashboard:	http://127.0.0.1:8265

[4]:
```python
ray.cluster_resources()
```

[4]:
```
{'GPU': 56.0,
 'node:100.112.169.99': 1.0,
 'CPU': 390.0,
 'memory': 134938159928.0,
 'object_store_memory': 59258226274.0,
 'accelerator_type:V100': 10.0,
 'node:100.76.72.113': 1.0,
 'node:100.121.31.109': 1.0,
 'node:100.115.112.62': 1.0,
 'node:__internal_head__': 1.0,
 'node:100.64.107.8': 1.0,
 'node:100.119.181.63': 1.0,
 'node:100.116.94.59': 1.0,
 'node:100.119.201.125': 1.0,
 'node:100.74.43.22': 1.0,
 'node:100.90.103.34': 1.0,
 'node:100.68.238.73': 1.0}
```

[ ]:

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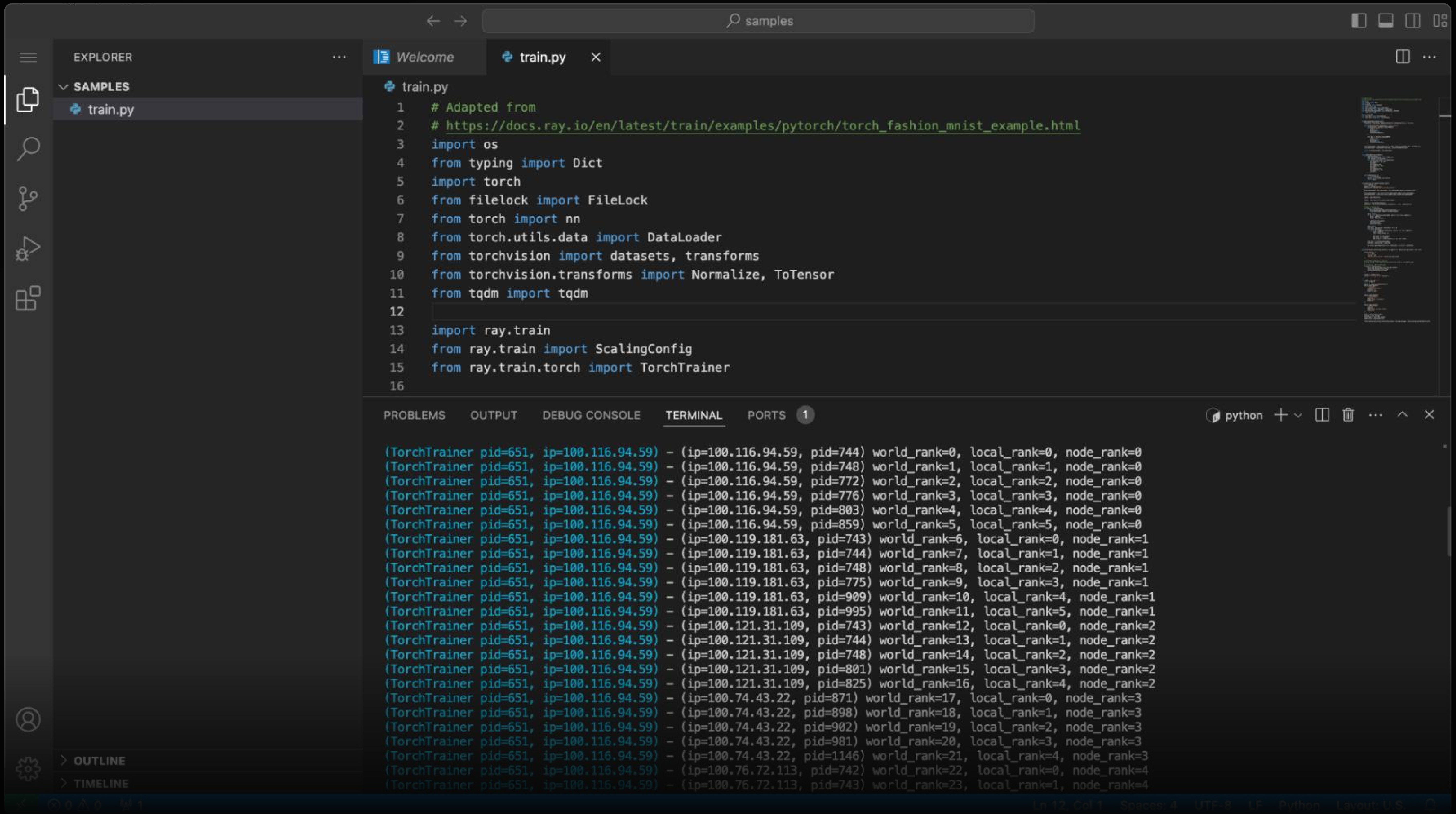
Distributed Training - Example with 56 GPUs

io.net



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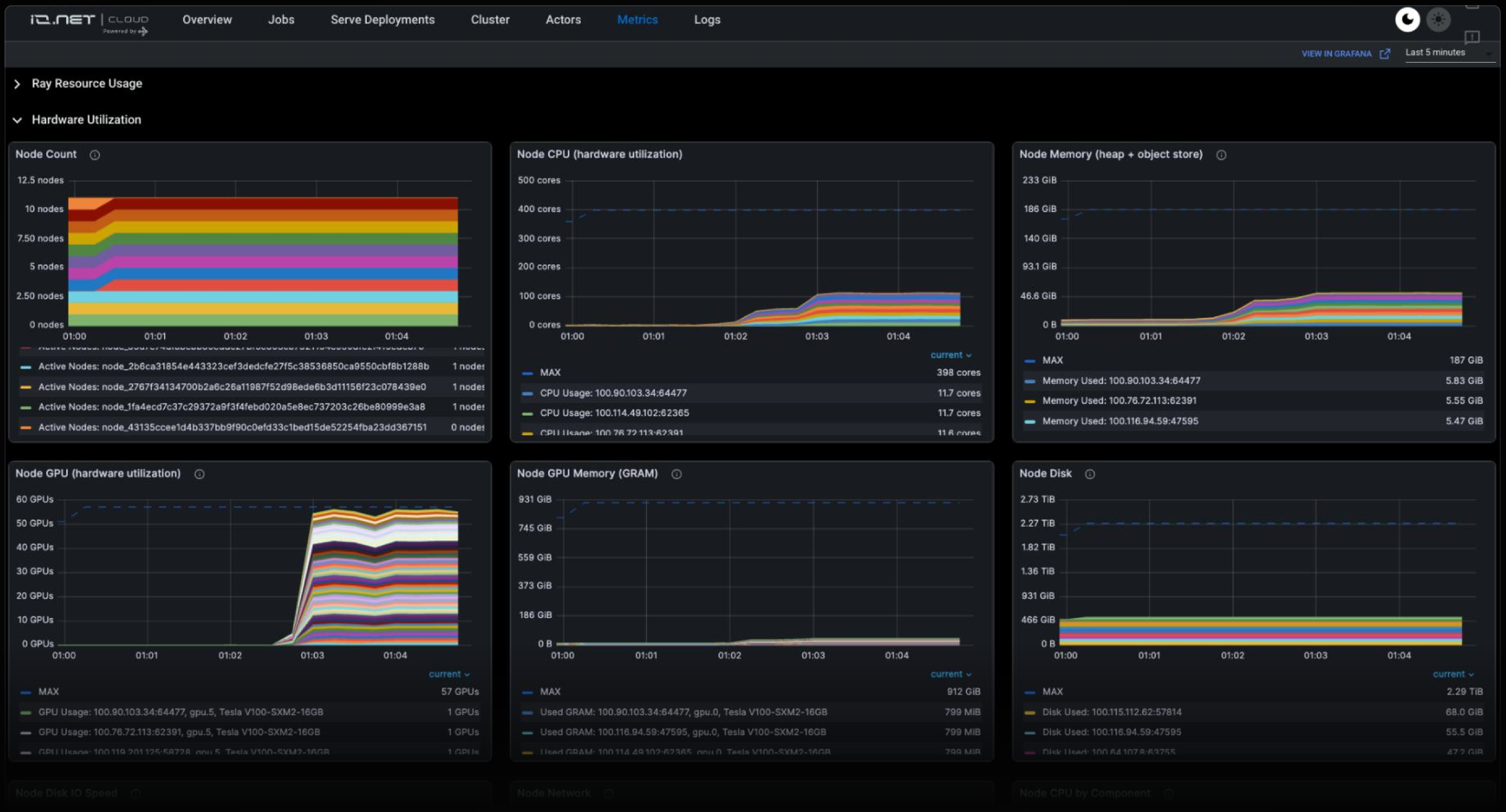
Distributed Training - Example with 56 GPUs



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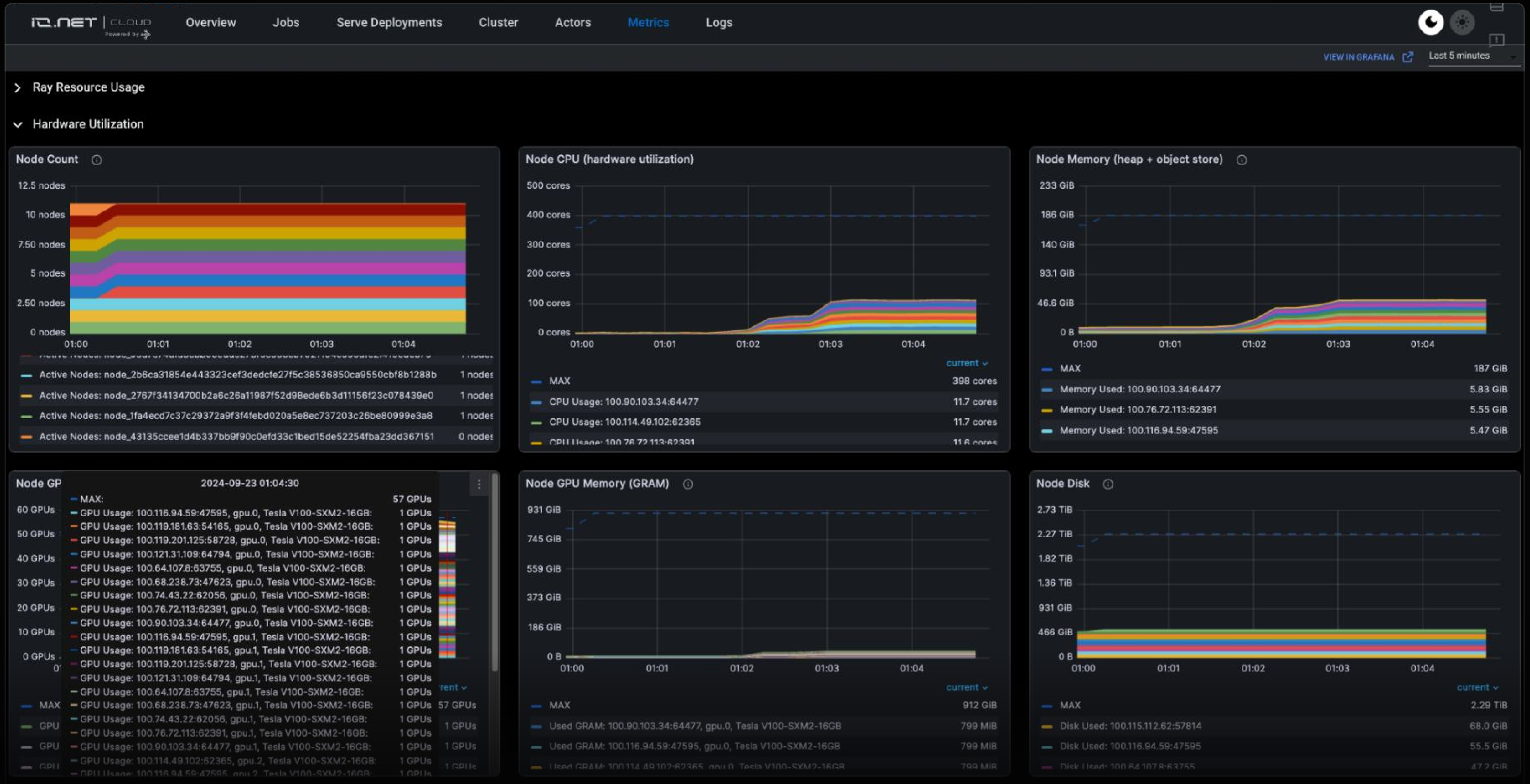
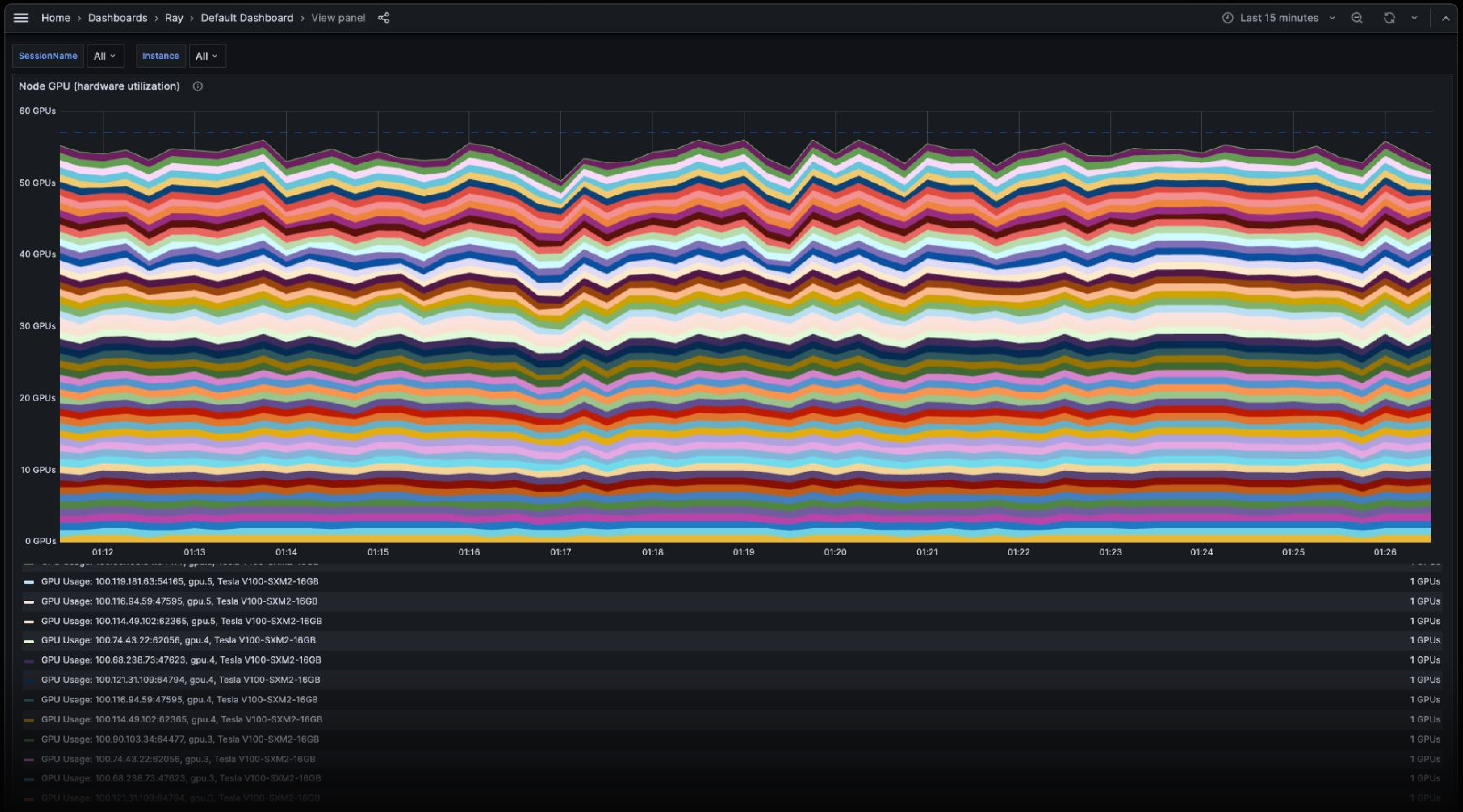
Distributed Training - Example with 56 GPUs



Distributed Training - Example with 56 GPUs



Ongoing Optimization Efforts

- IOG Foundation - Developer Tools
- Given a cluster request, location, type of workload, ...
  - Select the subset of nodes that satisfy conditions, besides reliability score
    - Maximum edge of the fully-connected subgraph (clique) is under the maximum latency allowed for the type of AI workload
- Minimize Communication Overhead
  - Be opinionated on trade-offs of more nodes vs communication overhead
  - Given a running cluster
    - Optimize resource allocation and task scheduling
  - Optimize topology / communication protocol for custom use cases
    - A fully-connected graph / network is not always needed

Ongoing Optimization Efforts

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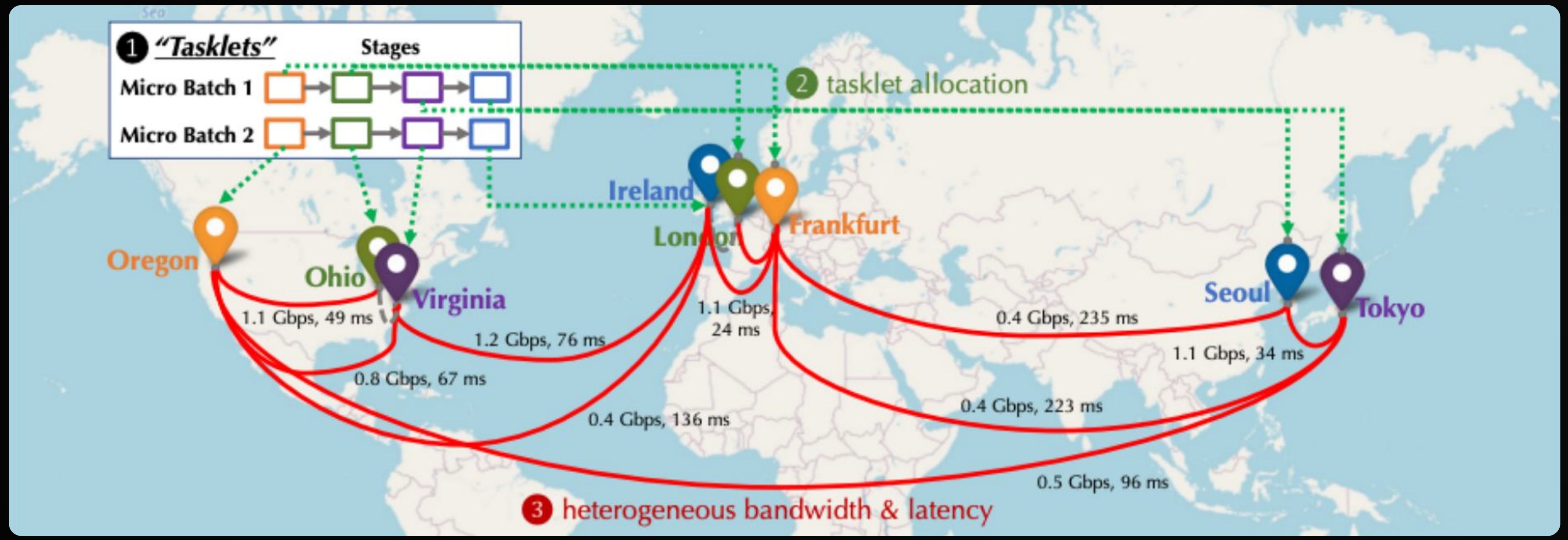
- **Open for Research Partnerships aligned with Decentralized Training / Inference, such as:**

  Distributed Inference and Fine-tuning of Large Language Models Over The Internet

  37th Conference on Neural Information Processing Systems (NeurIPS Dec 2023)
  Alexander Borzunov - HSE University, Yandex & Max Ryabinin - HSE University, Yandex & Artem Chumachenko Neiro.ai Dmitry Baranchuk Yandex & Tim Dettmers University of Washington & Younes Belkada Hugging Face Pavel Samygin Yandex School of Data Analysis & Colin Raffel Hugging Face

  https://arxiv.org/abs/2312.08361



  Decentralized Training of Foundation Models in Heterogeneous Environments

  Conference on Neural Information Processing Systems (NeurIPS Jun 2023).
  Binhang Yuan1∗ , Yongjun He1∗ , Jared Quincy Davis2 , Tianyi Zhang2 , Tri Dao2 , Beidi Chen34 , Percy Liang2 , Christopher Re2 , Ce Zhang1 1 ETH Zürich, Switzerland 2 Stanford University, USA 3 Carnegie Mellon University 4 Meta AI

  https://arxiv.org/abs/2206.01288

- **Make research reproducible**

  Provide blueprint for researchers to test and run novel algorithms in the field

Supporting Decentralized Training and Inference Research

- IOG Foundation sponsors Research with:
  - Infrastructure for testing and validating novel decentralized large-scale distributed training and inference
  - Partnerships with leading academic and industrial researchers to foster innovation in AI scalability

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Thank You.

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